FE515 2022A Assignment 2

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1.1

Question 1

1.1 Read data and convert first column to Date

The first column in the provided CSV is a row index, not calendar dates. If your original `JPM.csv` has real dates, this works unchanged. 1.1 Find the attached JPM.csv file. Use as.Date() function to change the first column to Date object.

```
jpm <- read.csv("FE515_hw2_JPM.csv")

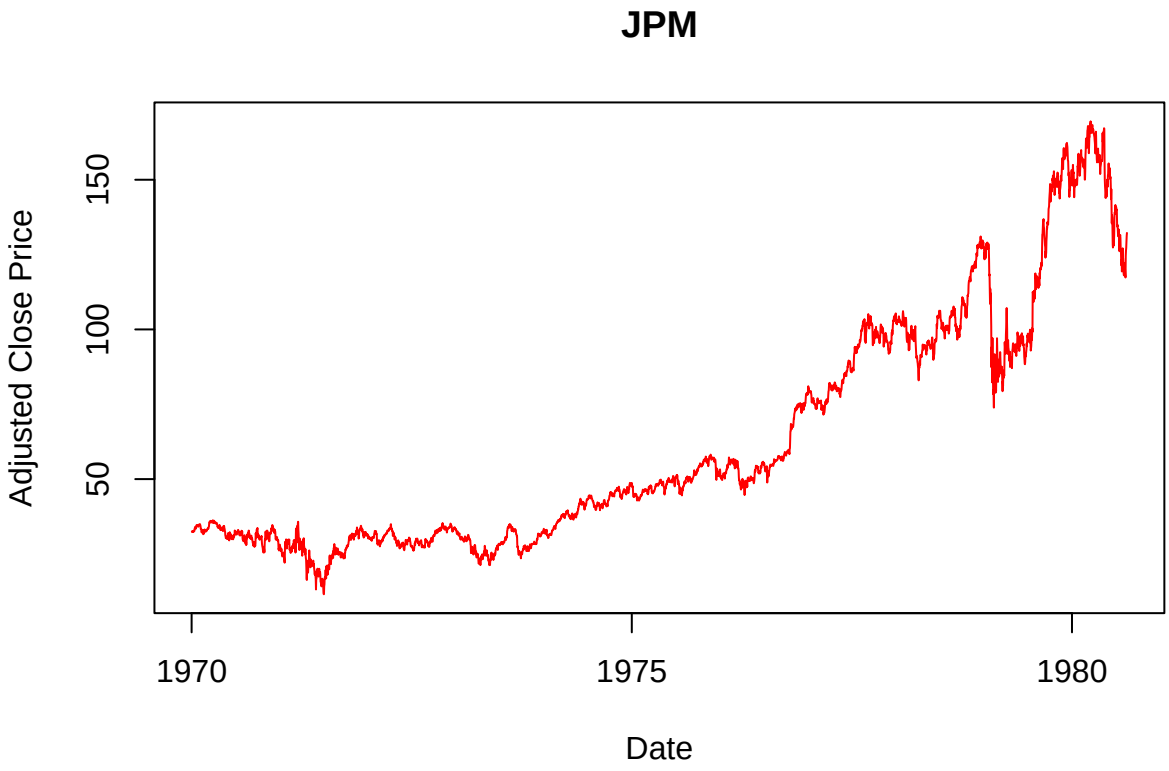
# adjust path if needed
jpm[, 1] <- as.Date(jpm[, 1])
colnames(jpm)[1] <- "Date"
head(jpm)
```

```
##         Date JPM.Open JPM.High JPM.Low JPM.Close JPM.Volume JPM.Adjusted
## 1 1970-01-02    48.00    48.37   47.59     48.07   14244700     32.52235
## 2 1970-01-03    48.05    48.55   47.75     48.19    9471500     32.60353
## 3 1970-01-04    48.17    48.25   47.63     47.79   10760500     32.33291
## 4 1970-01-05    47.57    48.06   47.32     47.95    8239200     32.44115
## 5 1970-01-06    47.90    48.11   47.36     47.75    9276700     32.30586
## 6 1970-01-07    47.47    48.12   47.44     48.10   15597000     32.54265
```

1.2

1.2 Plot the adjusted close price against the date object (i.e. date object as x-axis and close price as y-axis) in red line (require no points). Set the title as JPM, the label for x-axis as Date and the label for y-axis as Adjusted Close Price.

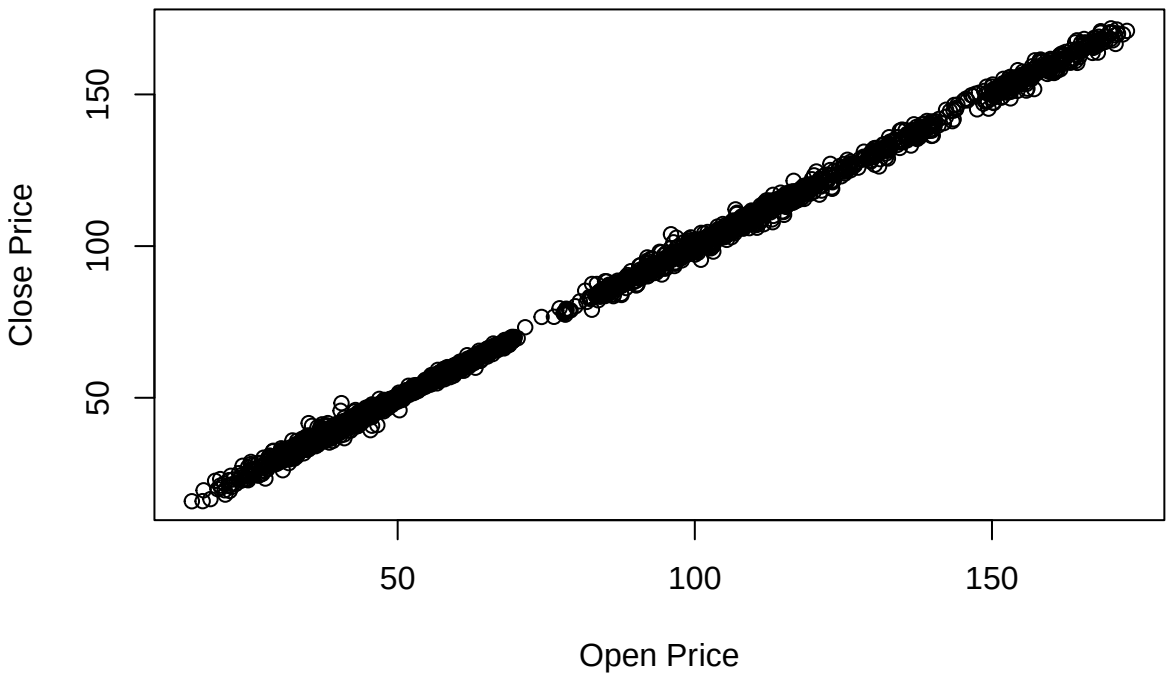
```
plot(jpm$Date, jpm$JPM.Adjusted, type = "l", col = "red",
     main = "JPM", xlab = "Date", ylab = "Adjusted Close Price")
```



1.3

1.3 Create a scatter plot of close price against open price (i.e. open prices as x-axis, and close prices as y-axis). Set the x label as "Open Price" and y label as "Close Price".

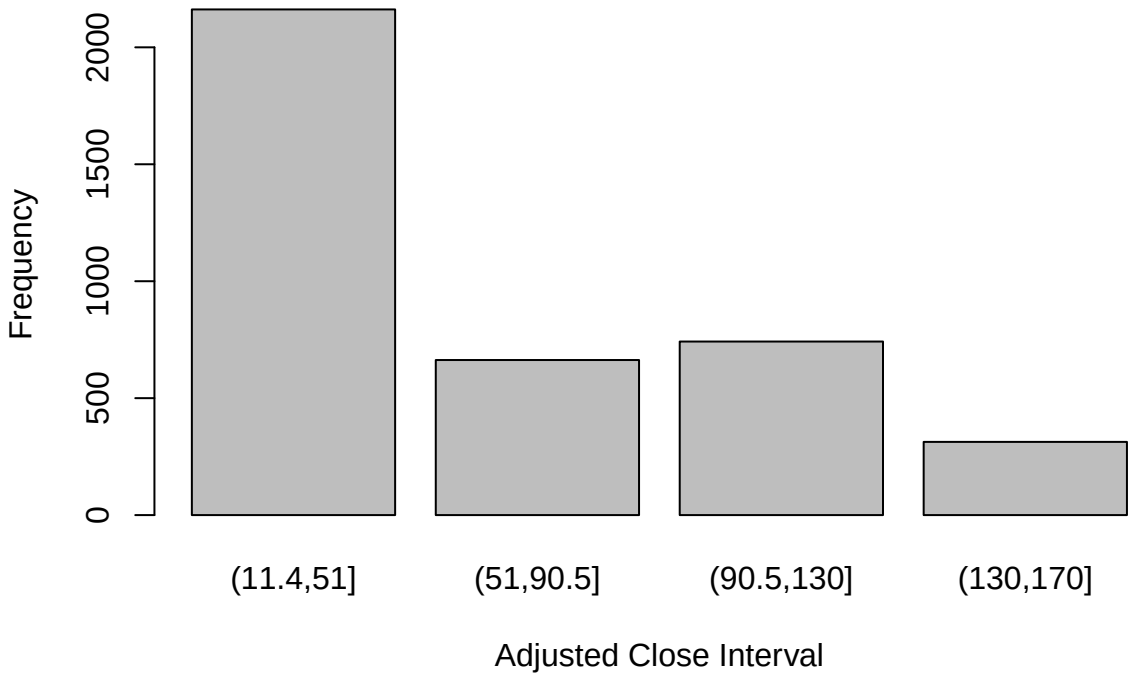
```
plot(jpm$JPM.Open, jpm$JPM.Close,
     xlab = "Open Price", ylab = "Close Price")
```



1.4

1.4 Use cut() function to divide adjusted close price into 4 intervals. Generate a barplot for the frequencies of these intervals.

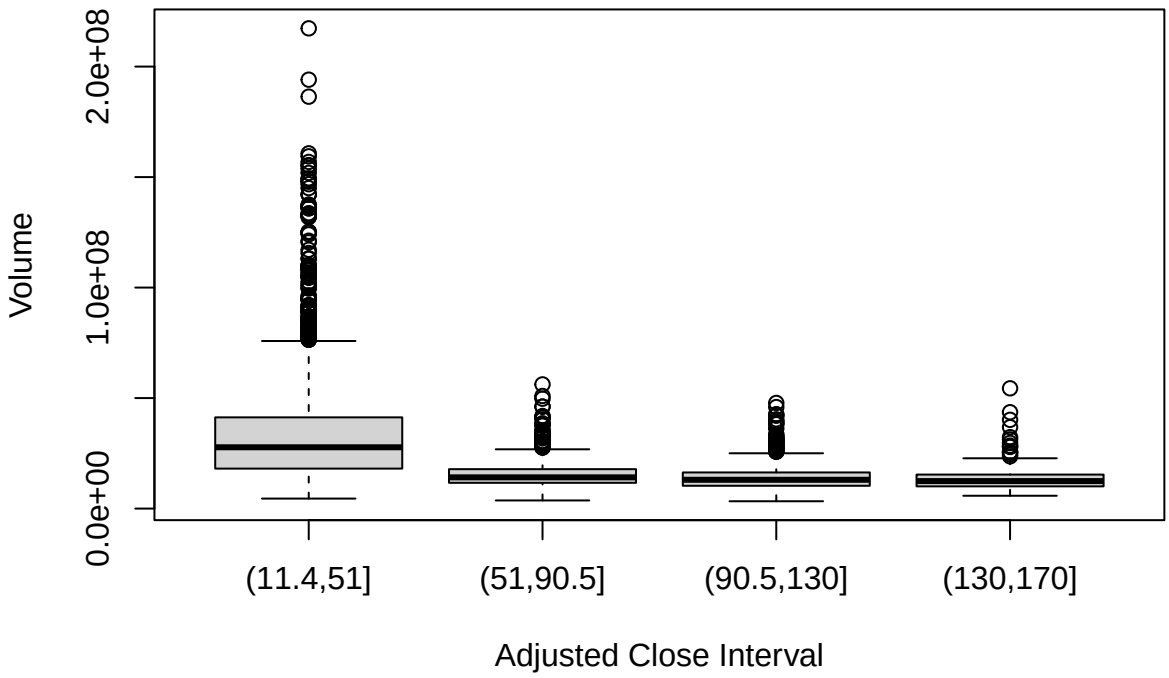
```
adj_int <- cut(jpm$JPM.Adjusted, breaks = 4)
barplot(table(adj_int), xlab = "Adjusted Close Interval", ylab = "Frequency")
```



1.5

1.5 Generate a boxplot of volume against the 4 intervals of adjusted close prices.

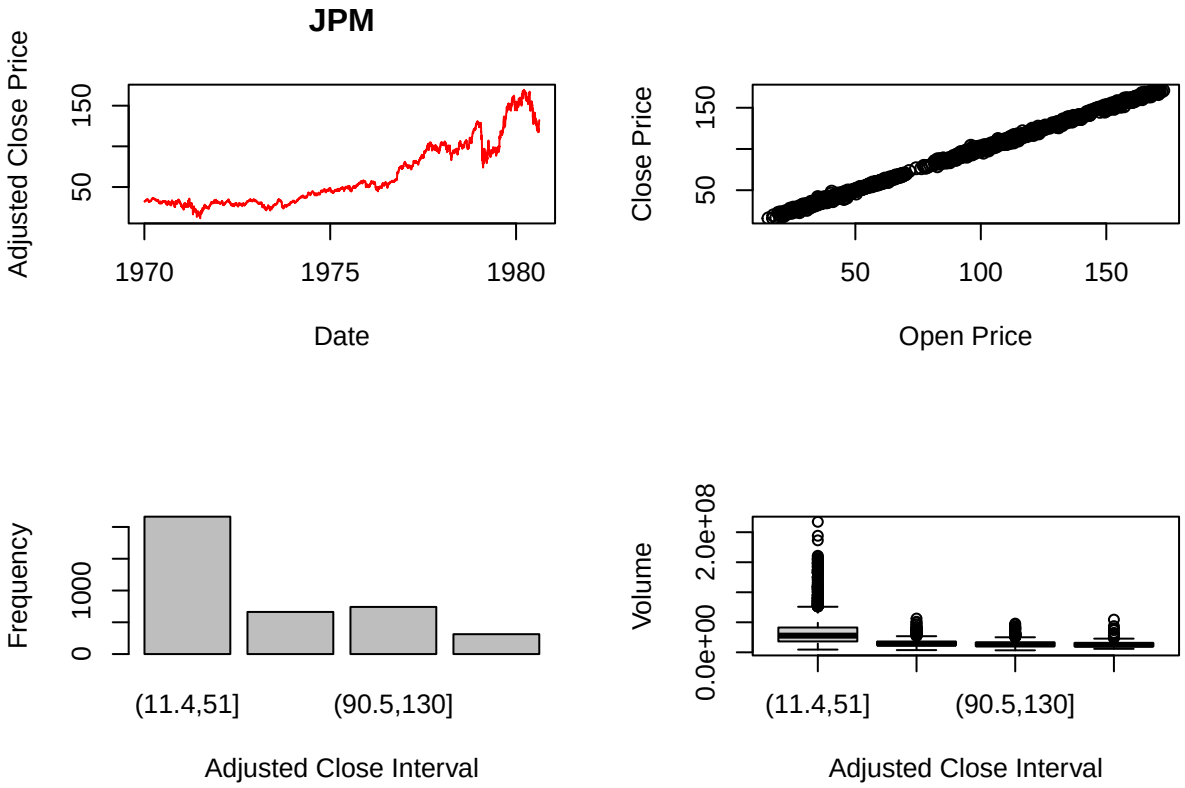
```
boxplot(jpm$JPM.Volume ~ adj_int,
        xlab = "Adjusted Close Interval", ylab = "Volume")
```



1.6

1.6 Use par() function to create a picture of 4 subplots. Gather the 4 figures from 1.2- 1.5 into ONE single picture. Please arrange the 4 subplots into a 2 by 2 frame, i.e. a frame consists of 2 columns and 2 rows.

```
par(mfrow = c(2, 2))
plot(jpm$Date, jpm$JPM.Adjusted, type = "l", col = "red",
     main = "JPM", xlab = "Date", ylab = "Adjusted Close Price")
plot(jpm$JPM.Open, jpm$JPM.Close,
     xlab = "Open Price", ylab = "Close Price")
barplot(table(adj_int), xlab = "Adjusted Close Interval", ylab = "Frequency")
boxplot(jpm$JPM.Volume ~ adj_int,
        xlab = "Adjusted Close Interval", ylab = "Volume")
```




```
par(mfrow = c(1, 1))
```

Question 2

Estimate the volume of the unit sphere (which is just $\frac{4\pi}{3}$) by simulation. we draw N points from three dimensional uniform distribution. In a 3-D co ordinate system, the position for the points inside the 1/8 unit sphere is given by

$$x^2 + y^2 + z^2 \leq 1, \quad x, y, z \geq 0.$$

$$\text{volume of } \frac{1}{8} \text{ unit sphere} = \frac{\text{number of inner points}}{\text{number of total points}} \times 1$$

Then we have the volume of the unit sphere.

$$\text{volume of unit sphere} = 8 \times \text{volume of } \frac{1}{8} \text{ unit sphere}$$

We can use the frequency of inner points to estimate the volume of 1/8 sphere.

```
set.seed(515)
N <- 1e6
x <- runif(N); y <- runif(N); z <- runif(N)
inner <- sum(x^2 + y^2 + z^2 <= 1)
```

```r
vol_eighth <- inner / N * 1      # eq. (2)
vol_sphere <- 8 * vol_eighth     # eq. (3)

cat("Estimated volume:", vol_sphere, "\n")
```

```
## Estimated volume: 4.18888
```

```r
cat("True volume     :", 4 * pi / 3, "\n")
```

```
## True volume     : 4.18879
```

Question 3

3.1 Implement a Linear Congruential Generator (LCG)

$$X_{n+1} = (aX_n + b) \bmod m$$

which generates pseudo-random number from uniform distribution using m = 244944,a = 1597,b = 51749. (Hint. See details in lecture 5 and example codes.)

```r
seed <- 1   # seed in global environment

LCG <- function(n){
  m <- 244944
  a <- 1597
  b <- 51749
  x <- rep(NA, n)
  x[1] <- (a * seed + b) %% m
  for(i in 1:(n-1)){
    x[i + 1] <- (a * x[i] + b) %% m
  }
  seed <<- x[n]
  return(x/m)
}
```

3.2 Chi-square (df = 10) sample

3.2 Use the LCG in the previous problem, generate 10000 random numbers from chi-square distribution with 10 degrees of freedom (i.e. df = 10), and assign to a variable.

```r
U <- LCG(10000)
X <- qchisq(U, df = 10)

summary(X)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.7883  6.7644  9.3320 10.0201 12.5361 43.4787
```

3.3 Histogram with 40 bins

Visualize the resulting sample from 3.2 using a histogram with 40 bins

```
hist(X, nclass = 40, main = "Chi-square (df = 10) via LCG", xlab = "X")
```

